

B13-CliniScan

Chest X-Ray AI Diagnostics

MILESTONE 3 REPORT

Hyperparameter Tuning, Advanced Augmentation & Model Improvement

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Project	B13-CliniScan
Milestone	Milestone 3 — Model Enhancement
Platform	CPU — AMD Ryzen 3 3250U, 8GB RAM

1. Introduction

Milestone 3 builds on the Milestone 2 baseline by applying systematic hyperparameter tuning, advanced data augmentation, transfer learning with alternate backbones, and comprehensive model evaluation. The goal is to improve both classification accuracy and detection performance on chest X-ray images.

This document covers all six Milestone 3 tasks:

- Task 1: Hyperparameter Tuning — optimizer, learning rate, epochs
- Task 2: Advanced Data Augmentation — Albumentations pipeline
- Task 3: Transfer Learning — alternate backbones, layer unfreezing
- Task 4: Full Validation Evaluation — AUC, F1, Precision, Recall
- Task 5: Grad-CAM Heatmaps — model interpretability
- Task 6: Error Analysis — false negatives, false positives, patterns

2. Milestone 2 Baseline

2.1 Classification Baseline

The M2 classification model used EfficientNet-B0 pretrained on ImageNet, fine-tuned for multi-label classification across 14 chest X-ray disease classes.

Parameter	Value	Notes
Model	EfficientNet-B0	ImageNet pretrained
Optimizer	Adam	Default settings
Learning Rate	1e-3	Fixed LR
Epochs	10	M2 training
Batch Size	8 (CPU)	CPU constraint
Val AUC-ROC	0.8858	M2 best result
Val F1-Score	0.3279	Macro average

2.2 Detection Baseline

The M2 detection model used YOLOv8s trained for 30 epochs with YOLO-format bounding box annotations on the NIH chest X-ray dataset.

Metric	M2 Value	Notes
Model	YOLOv8s	Small variant
mAP@50	0.0658	Main detection metric
mAP@50-95	0.0286	Strict IoU metric
Precision	0.3793	TP / (TP+FP)
Recall	0.0821	Low — needs improvement

3. Task 2 — Advanced Data Augmentation

The M2 pipeline used basic torchvision transforms. For M3, the pipeline was rebuilt using Albumentations with medically safe augmentations chosen specifically for chest X-ray data.

3.1 Augmentation Comparison Table

Augmentation	M2	M3	Medical Reason
Horizontal Flip	Yes	Yes p=0.5	Left-right symmetry valid in chest X-rays
Rotation +/-10 deg	Yes	Yes p=0.5	Mild patient positioning variation between scans
Brightness/Contrast	Yes	Yes p=0.5	Exposure variation between different X-ray machines
CLAHE	No	NEW p=0.4	Local contrast — reveals subtle findings like early nodules
Gaussian Noise	No	NEW p=0.3	Simulates scanner noise for better robustness
Random Crop/Zoom	No	NEW p=0.3	Scale invariance — lesions appear at different zoom levels
Vertical Flip	No	DISABLED	Would invert anatomy — heart above diaphragm
Perspective Warp	No	DISABLED	Distorts lung shape — misleads spatial feature learning
Color Jitter	Yes	DISABLED	Chest X-rays are grayscale — color shifts have no meaning

3.2 Why CLAHE is Critical for Chest X-Rays

CLAHE (Contrast Limited Adaptive Histogram Equalization) divides the image into small tiles and equalizes each tile independently. This enhances local contrast in different anatomical regions — ribs, lung fields, cardiac silhouette — independently. Standard global histogram equalization would over-amplify noise in already high-contrast regions. CLAHE's clip limit parameter prevents this, making it ideal for medical images where subtle texture differences indicate pathology.

4. Task 1 + 3 — Hyperparameter Tuning & Transfer Learning

Four classification experiments were conducted covering all M3 Task 1 and Task 3 requirements. All experiments used the M3 Albumentations pipeline and CosineAnnealing LR scheduler. Training ran on CPU with batch size 8 and 12 epochs per experiment.

4.1 Experiment Configuration

Experiment	Backbone	LR	Optimizer	Epochs	Unfreeze	M3 Task
Exp1_Adam	EfficientNet-B0	1e-3	Adam	12	0	Task 1 - LR ref
Exp2_AdamW	EfficientNet-B0	5e-4	AdamW	12	0	Task 1 - Optimizer
Exp3_SGD	EfficientNet-B0	1e-2	SGD+momentum	12	0	Task 1 - SGD
Exp4_DenseNet121	DenseNet121	1e-3	Adam	12	0	Task 3 - Backbone

4.2 Results

Experiment	Best AUC	vs M2 Baseline	Result	Notes
Exp2_AdamW (BEST)	0.9213	+0.0355	IMPROVED	Best optimizer
Exp1_Adam	0.9104	+0.0246	IMPROVED	Good baseline
Exp3_SGD	0.9094	+0.0236	IMPROVED	Needs more epochs
Exp4_DenseNet121	0.8304	-0.0554	LOWER	Needs 30+ epochs
Baseline M2 (ref)	0.8858	---	----	Adam 10 epochs

4.3 Why AdamW Beat Adam

Standard Adam optimizer applies weight decay by adding it directly to the gradient — this incorrectly scales the decay by the adaptive learning rate and leads to less effective regularization. AdamW decouples weight decay from the gradient update, applying it directly to the weights after the gradient step. This produces cleaner regularization, prevents overfitting on small datasets like the 3000-image NIH subset, and is now the recommended default for fine-tuning pretrained models.

5. Detection Hyperparameter Tuning

Four YOLOv8 detection experiments were designed to improve the M2 mAP@50 of 0.0658. Experiments cover backbone size upgrade, learning rate reduction, confidence threshold tuning, and SGD optimizer comparison. All use medically safe augmentations only.

5.1 Experiment Configuration

Name	Model	LR	Conf	IoU	Epochs	Optimizer	M3 Task
D1_YOLOv8m	YOLOv8m	0.001	0.25	0.50	15	AdamW	Task 3 Backbone
D2_LowLR	YOLOv8s	0.0005	0.25	0.50	15	AdamW	Task 1 LR
D3_LowThresh	YOLOv8s	0.001	0.15	0.45	15	AdamW	Task 1 Threshold
D4_SGD	YOLOv8s	0.01	0.25	0.50	15	SGD	Task 1 Optimizer

5.2 Results

Model	mAP@50	mAP@50-95	Precision	Recall	vs M2
D1_YOLOv8m	0.0463	0.0202	0.6951	0.0527	- 0.0195
D2_LowLR (BEST mAP)	0.0592	0.0251	0.3683	0.0725	- 0.0066
D3_LowThresh	0.0574	0.0265	0.1749	0.0624	- 0.0084
D4_SGD	0.0446	0.0173	0.5503	0.0436	- 0.0212
Baseline YOLOv8s M2	0.0658	0.0286	0.3793	0.0821	ref

5.3 Analysis of Detection Results

All four M3 detection experiments scored below the M2 baseline mAP@50 of 0.0658. This is a known challenge in medical object detection — the M2 model was trained for 30 epochs while all M3 experiments ran for only 15 epochs due to CPU time constraints. Detection models need significantly more training time than classification models to converge on complex multi-class localization tasks.

However the experiments revealed important findings about the precision-recall tradeoff:

- **The larger YOLOv8m backbone achieved precision of 0.6951 — nearly double the M2 baseline of 0.3793. This means when YOLOv8m does make a detection, it is much more likely to be correct. With more training epochs it would likely also improve recall.** D1_YOLOv8m precision:
- **Halving the learning rate to 5e-4 gave the best overall mAP@50 of 0.0592 and the best recall of 0.0725. Lower LR allows finer weight updates which helps with the subtle bounding box regression task.** D2_LowLR best mAP:

- Lowering the confidence threshold to 0.15 increased recall (more boxes predicted) but dramatically reduced precision to 0.1749. This is the expected precision-recall tradeoff — lower threshold = more detections including false positives. D3_LowThresh tradeoff:
- SGD achieved the lowest mAP@50 of 0.0446 in 15 epochs, confirming that SGD requires significantly more training iterations than AdamW to converge on detection tasks. D4_SGD convergence:

5.4 Medically Safe Detection Augmentations

Augmentation	Setting	Medical Justification
Horizontal Flip	fliplr=0.5	Left-right symmetry is valid — chest organs have known lateralization but flipping is acceptable
Rotation	degrees=10	Small patient positioning variation between scans — large rotation would move organs unrealistically
Scale	scale=0.1	Variation in detector-to-patient distance changes apparent organ size
Brightness only	hsv_v=0.1	X-ray exposure varies by machine settings — brightness adjustment only, no hue or saturation
Vertical Flip	DISABLED	Would place heart below diaphragm — completely invalid anatomy
Mosaic / Mixup	DISABLED	Combines multiple patients into one image — clinically impossible and misleads detector
Perspective / Color	DISABLED	Perspective distorts lung shape. Color shift is meaningless for grayscale X-rays

6. Task 5 — Grad-CAM Interpretability

Grad-CAM (Gradient-weighted Class Activation Mapping) was applied to the best M3 classification model to generate visual explanations for each prediction. This is critical for medical AI validation — clinicians need to verify the model attends to clinically relevant regions, not background artifacts.

6.1 How Grad-CAM Works

Grad-CAM computes the gradient of the target class score with respect to the final convolutional layer feature maps. These gradients are global-average-pooled to produce importance weights. A weighted sum of the feature maps is computed and passed through a ReLU — producing a heatmap where red areas indicate regions most important for the prediction.

6.2 Target Layers

Backbone	Grad-CAM Layer	Why
EfficientNet-B0	model.features[-1]	Last MBConv block — highest semantic spatial resolution
DenseNet121	model.features[-2]	Last dense block before transition layer

6.3 Outputs Generated

- gradcam_grid.png — 12 test images shown side by side with top-class heatmaps (3 rows x 8 columns)
- gradcam_multiclass.png — 3 images each visualized with heatmaps for 5 disease classes: Aortic enlargement, Cardiomegaly, Nodule/Mass, Pleural effusion, Pneumothorax
- gradcam/ folder — 12 individual heatmap PNG files for detailed review in the report

7. Task 6 — Error Analysis

A systematic error analysis was performed on the complete test set using the best M3 model (Exp2_AdamW, AUC=0.9213). Every test image was evaluated and categorized.

7.1 Error Categories and Clinical Impact

Category	Definition	Clinical Impact
False Negative	Disease present, model missed it	MOST CRITICAL — patient with disease is cleared without treatment
False Positive	No disease, model predicted it	Causes unnecessary follow-up tests and patient anxiety
Correct Prediction	Prediction matches ground truth	No action required

7.2 Error Analysis Outputs

- error/false_negatives.png — top 6 worst false negative cases with ground truth and missed class annotations
- error/false_positives.png — top 6 false positive cases
- error/class_error_breakdown.png — horizontal bar chart of miss frequency per disease class

- error/confidence_dist.png — histogram comparing confidence on correct predictions vs missed ones
- error/summary.json — counts, accuracy, and most problematic classes in machine-readable format

8. Saved Model Files

All trained models are saved as PyTorch .pt files (pickle format internally). Each file contains the complete trained weights reflecting the specific hyperparameter configuration used. The best model (Exp2_AdamW) is the final M3 classification model.

File	Contents	Size	Result
Exp1_Adam_best.pt	EfficientNet-B0 + Adam 1e-3	~22 MB	AUC=0.9104
Exp2_AdamW_best.pt (FINAL)	EfficientNet-B0 + AdamW 5e-4	~22 MB	AUC=0.9213 BEST
Exp3_SGD_best.pt	EfficientNet-B0 + SGD 1e-2	~22 MB	AUC=0.9094
Exp4_DenseNet121_best.pt	DenseNet121 + Adam 1e-3	~95 MB	AUC=0.8304

9. Complete Deliverables

9.1 Classification — milestone3_cls/

File / Folder	Description
plots/aug_comparison.png	3 x 4 augmentation side-by-side preview
plots/exp_all_metrics.png	AUC + F1 + Precision + Recall curves for all experiments
plots/auc_bar.png	AUC comparison bar chart vs M2 baseline
plots/per_class_metrics.png	Per-class F1 and AUC for all 14 disease classes
gradcam_grid.png	12-image Grad-CAM heatmap grid
plots/gradcam_multiclass.png	3 images x 5 disease class heatmaps
gradcam/ (12 PNG files)	Individual heatmap files per test image
error/false_negatives.png	Top 6 missed disease cases
error/false_positives.png	Top 6 false alarm cases

error/class_error_breakdown.png	Per-class FN/FP frequency bar chart
error/confidence_dist.png	Confidence histogram TP vs FN
experiment_results.csv	All 4 experiment metrics in CSV
report.txt	Full text report with all metrics and findings

9.2 Detection — milestone3_det/

File / Folder	Description
D1_YOLOv8m/ to D4_SGD/	weights/best.pt + results.csv + auto-plots
correct_detections.png	8 correct detection examples with bounding boxes
missed_detections.png	8 failure cases where model found nothing
plots/m2_vs_m3.png	Side-by-side M2 vs M3 on same 8 test images
plots/threshold_analysis.png	Precision/Recall/mAP across 8 confidence thresholds
det_results.csv	All 4 detection experiment metrics

10. Summary

Metric	M2 Baseline	M3 Result
Classification AUC	0.8858	0.9213 (+3.55%)
Best Cls Optimizer	Adam	AdamW (decoupled WD)
Detection mAP@50	0.0658	0.0592 — D2_LowLR best
Augmentation	torchvision basic	Albumentations + CLAHE + Noise + Crop
Interpretability	None	Grad-CAM heatmaps (12 images)
Error Analysis	None	FN/FP grids + class breakdown + confidence dist